Applications of Integration to *Problems*

Problems = Area between curves
Volume of solids
Area in polar coordinates
Work
Fluid force
Moments
Centroids

Process

a. Subdivide the *problem* into smaller *subproblems*

$$P = P_1 \cup P_2 \cup P_3 \cup \cdots \cup P_n$$

b. Formulate an approximation for the value of each *subproblem*

$$Value(P_k) \approx V_k = F(x_k^*)\Delta x_k$$

c. Sum up the approximate values (form a Riemann Sum)

$$V_1 + V_2 + V_3 + \cdots + V_n =$$

$$F(x_1^*)\Delta x_1 + F(x_2^*)\Delta x_2 + F(x_3^*)\Delta x_3 + \cdots + F(x_n^*)\Delta x_n =$$

$$\sum_{k=1}^{n} F(x_k^*)\Delta x_k = RS$$

d. Limiting Case as $\|\Delta_k\| \to 0$

$$RS = \sum_{k=1}^{n} F(x_k^*)\Delta x_k \to \int_a^b F(x)dx$$